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Fostering inclusive learning through serious gaming to enhance sustainable irrigation investments along sand rivers in Zimbabwe and Kenya

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ABSTRACT

Smallholder farmers across sub-Saharan Africa increasingly invest in small-scale irrigation to improve agricultural productivity and enhance livelihoods. Unlike conventional top-down irrigation development, these investments are typically incremental and adaptive, reflecting changing environmental and socio-economic conditions. However, limited empirical evidence exists on how farmers make irrigation investment decisions in practice, particularly in sand river systems where groundwater stored in sandy riverbeds provides a critical seasonal water source. This study presents the Sand River Game, a board-based serious game designed to simulate irrigation investment decision-making and facilitate dialogue on irrigation development pathways. The game was tested with irrigating farmers along the Tuli River in Zimbabwe and the Enguli River in Kenya, two contexts where farmer-led irrigation development is expanding. Gameplay revealed diverse irrigation investment pathways shaped by contextual factors such as access to credit, market opportunities, and perceptions of risk. Farmers experimented with irrigation technologies, crop choices, and risk-management strategies. Post-game reflections highlighted trade-offs related to energy costs, water availability, and collective marketing. The findings demonstrate how serious gaming can support experiential learning and dialogue around irrigation investment decisions and provide a practical tool for grounding the concept of Adaptive Investment Pathways in smallholder irrigation development along sand rivers.

Key words: Adaptive investment pathways, Farmer-led irrigation development, Participatory learning, Sand rivers, Serious games, Smallholder farmers

HIGHLIGHTS

- Serious game developed to explore farmer-led irrigation investment pathways.
- Sand River Game links Adaptive Investment Pathways with experiential learning.
- Farmers tested irrigation, crop, and risk-management strategies under uncertainty.

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- Context shaped investment decisions differently in Zimbabwe and Kenya.
- Serious gaming stimulated dialogue on sustainable sand river irrigation development.

1. INTRODUCTION

Agricultural production in sub-Saharan Africa (SSA) remains highly vulnerable to semi-arid conditions and variable rainfall, which constrain crop productivity and contribute to persistent food insecurity and poverty for millions of agriculture-dependent households (United Nations 2015; Giller *et al.* 2021). Irrigation has long been recognized as a key strategy for stabilizing agricultural production and strengthening climate resilience in the region. However, irrigation development in SSA remains limited: only about 6% of cultivated land is equipped for irrigation, and approximately 20% of this area remains unused (Wiggins & Lankford 2019; Nhamo *et al.* 2024). Large-scale, centralized irrigation schemes continue to dominate irrigation policy and investment strategies, yet many of these schemes underperform due to high capital costs, misalignment with local hydrological conditions, and weak responsiveness to the socio-economic realities of smallholder farmers (Higginbottom *et al.* 2021; Kikuchi *et al.* 2021). As a result, irrigation development in many countries follows a recurring 'build–neglect–repair' cycle in which infrastructure deteriorates and requires repeated rehabilitation (Lankford *et al.* 2016; Waalewijn *et al.* 2020).

In response to these limitations, farmer-led irrigation development (FLID) has emerged as an important pathway for accelerating irrigation in many parts of SSA, where farmers independently initiate, finance, and manage irrigation systems often with minimal external support (Veldwisch *et al.* 2019). These systems typically develop incrementally as farmers make a series of small investments over time in water abstraction technologies, irrigation methods, and crop production. Such investments are shaped by evolving environmental conditions, financial constraints and opportunities, market conditions, and livelihood priorities (Prasad *et al.* 2023). Unlike centralized irrigation schemes, which require large upfront investments and fixed infrastructure, farmer-led irrigation systems tend to be more flexible and adaptive, both by choice and necessity, allowing farmers to experiment with different strategies and gradually expand their irrigation activities. However, this flexibility can also come with important limitations, including reliance on limited financial resources, exposure to risk, uneven access to water and technologies, and the potential for uncoordinated development that may undermine long-term sustainability.

Ephemeral sand rivers represent important water sources supporting farmer-led irrigation in semi-arid regions. These rivers contain thick deposits of sand that store volumes of shallow groundwater following episodic flood events, creating shallow aquifers that can be accessed through scoop wells and small pumps (Mansell & Hussey 2005; Chauruka *et al.* 2023). In many parts of southern and eastern Africa, farmers increasingly rely on sand rivers for small-scale irrigation because they are reliable water sources during dry seasons. However, sand river aquifers function as common-pool resources and individual irrigation investments can influence water availability for other users. As irrigation expands, farmers must therefore make decisions that balance private investment objectives with shared resource constraints, creating potential trade-offs between productivity, equity, and sustainability (Duker *et al.* 2025).

The Adaptive Investment Pathways (AdIP) framework has been proposed as a useful lens for understanding how irrigation development occurs under such uncertain and dynamic conditions (Prasad *et al.* 2023, 2024). Rather than assuming that irrigation development follows a predetermined plan, the AdIP approach emphasizes that irrigation systems evolve through sequential decisions made by farmers and other actors as they respond to changing environmental, economic, and institutional conditions. Early investments influence the range of future options available, creating branching pathways of development over time. While the AdIP framework provides a valuable conceptual perspective, there is limited empirical evidence on how farmers navigate these investment

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pathways in practice. This gap is particularly pronounced in sand river irrigation systems, where farmers make investment decisions that may have implications for other irrigators and water users, yet these interdependencies are not always explicitly considered or well understood. Understanding how farmers make investment decisions while balancing their own objectives with those of other water users is critical for assessing irrigation development dynamics across the landscape. This applies not only to investment choices in sand river irrigation systems but also to broader policy and development domains.

Conventional research methods, such as surveys, provide partial insights into the decision-making processes. This limitation arises because farmers often make investment decisions through iterative experimentation, learning from both successes and failures. Capturing such dynamics requires participatory approaches that allow stakeholders to explore different strategies and reflect on their consequences in a structured setting. Serious games offer one such approach. Serious games are interactive simulations designed not primarily for entertainment but for learning, exploration, and decision support (Aubert *et al.* 2018; Akhtar *et al.* 2020). In natural resource management, serious games have increasingly been used to help stakeholders understand complex socio-ecological systems, experiment with alternative strategies, and reflect on the consequences of their decisions (Creutzig & Kapmeier 2020; Hallinger *et al.* 2020). By simplifying real-world dynamics into structured gameplay, serious games enable participants to test decisions in a low-risk environment while revealing trade-offs among competing objectives.

In the context of irrigation development, serious games can be particularly valuable for exploring how farmers allocate resources among competing investment options such as irrigation technologies, crop choices, and risk-management strategies (Wamucii *et al.* 2024). Simulation-based and role-playing game formats allow participants to experiment with irrigation infrastructure investments, respond to environmental shocks, and negotiate resource use with other players. These interactions can reveal how investment pathways emerge and how contextual factors influence decision-making processes. Despite the growing use of serious games in water resource management, empirical studies applying serious gaming to FLID and AdIP have not been reported, in SSA, or to our knowledge, elsewhere.

This study addresses this gap by developing and testing the Sand River Game, a board-based serious game designed to simulate irrigation investment decisions in sand river systems by individual smallholder farmers. The game creates a simplified representation of a farming environment in which participants make decisions regarding irrigation technologies, crop production, and complementary livelihood investments while responding to climate variability, market fluctuations, and financial constraints and opportunities. Through gameplay and structured reflection sessions, the simulation enables farmers to explore alternative irrigation development strategies and discuss their potential implications for productivity, risk management, and resource governance.

The game was implemented with irrigating farmers along two sand rivers in semi-arid regions: the Tuli River in Gwanda District, Zimbabwe, and the Enguli River in Makueni County, Kenya. These locations represent contrasting contexts of sand river irrigation development, with sand rivers in Zimbabwe tending to be underutilized compared to those in Kenya (Duker *et al.* 2020). Comparing these contexts provides an opportunity to examine how environmental, institutional, and economic conditions influence, in a game-playing context, irrigation investment pathways and how participatory simulations affect the content and quality of the discussions. This study contributes empirically to the literature on FLID and AdIP by demonstrating how serious gaming can be used to explore irrigation investment trajectories, reveal trade-offs in irrigation decision-making, and stimulate collective reflection among farmers operating under contrasting institutional and macro-economic conditions. The study addresses two research questions:

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1. How does the design and structure of the Sand River Game support learning, reflection, and dialogue among smallholder farmers regarding irrigation investment decisions?
2. How do differences in local context shape discussions, decision-making processes, and irrigation investment pathways?

By integrating serious gaming with the AdIP framework, this research contributes new empirical insights into how smallholder farmers make irrigation investment decisions in sand river systems. More broadly, the study demonstrates how participatory simulation tools can support dialogue and learning around irrigation development pathways in contexts where irrigation expansion is driven primarily by individual farmer initiatives rather than centralized planning.

2. METHODS

2.1. Study approach

This study employed a participatory serious gaming approach using a board-based simulation game to simulate farmer-led irrigation investments. Data was collected through gameplay sessions, participant observation, and facilitated reflection discussions with irrigating farmers immediately after playing the game. The approach was designed to address two key research questions presented in Section 1 of the paper.

The participatory serious gaming approach was selected because conventional methods such as surveys and semi-structured interviews are limited in their ability to capture iterative and sequential investment decision-making processes under uncertain conditions. Surveys often provide cross-sectional representations of farmer behaviour at a single point in time, while semi-structured interviews frequently rely on retrospective recall, which may inadequately represent evolving decision making processes and adaptive responses to changing conditions and shocks (Wollburg *et al.* 2021; Godlonton *et al.* 2024). In contrast, serious gaming enables participants to experiment with alternative strategies, negotiate decisions collectively, and observe the consequences of sequential investment choices within a structured but low-risk environment (Valkering *et al.* 2013; Speelman *et al.* 2014). This makes the approach particularly suitable for examining irrigation investment pathways within the AdIP framework, where development occurs through cumulative and adaptive decisions over time.

The study was implemented in three stages: (i) game design and development, (ii) testing and refinement, and (iii) implementation of gameplay sessions with irrigation farmers in Zimbabwe and Kenya. The game design was grounded in extensive fieldwork and prior research in both study areas, ensuring that key elements such as irrigation practices, risks, and investment options reflected locally relevant conditions and resonated with participating farmers. The description of the stages is preceded by an overview of the study sites.

2.2. Study sites

Gameplay sessions were conducted with farmers along two sand rivers in semi-arid regions of Zimbabwe and Kenya: a 12.5 km stretch of the Tuli River in Wards 16 and 17 of Gwanda District, Zimbabwe, and a 2 km stretch of the Enguli River in Kasikeu Ward, Kilome Sub-County, Makueni County, Kenya. These two sites were selected to capture contrasting contexts of sand river irrigation development, allowing the study to examine how different environmental, socio-economic and institutional settings influence farmer investment decisions and reflections.

Both regions experience low and highly variable rainfall – less than 450 mm per year in Gwanda District (Mupangwa *et al.* 2023) and 350–450 mm per year in Makueni County (Government of Makueni County 2019). These climatic conditions make rain-fed agriculture unreliable and contribute to diversified livelihood

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strategies that include livestock rearing, smallholder crop production, artisanal gold mining in Zimbabwe, and sand harvesting in Kenya.

Smallholder irrigation occurs in both sites but differs in scale and intensity. Along the Tuli River in Zimbabwe, five collective irrigation schemes exist, : Mankonkoni, Sebasa, Rustlers Gorge, Guyu Chelesa, and Tuli Shumba – covering a combined area of approximately 240 ha and benefitting about 420 households, with an average irrigated area of 0.6 ha per farmer (Chauruka *et al.* 2023, unpublished field notes). These, however, perform poorly and cater for a small proportion of the population (Moyo *et al.* 2016). Other farmers undertake individual irrigation, which is the focus of this article, and use water stored in sand river systems, relying on manual means for abstracting water, and petrol and solar pumps or a combination of these (Chauruka *et al.* 2023). Along the Enguli River in Kenya, individual irrigation is more intensively practiced, with farmers relying on sand river water to cultivate vegetables, fruits, and staple crops such as maize and beans (Duker *et al.* 2025). Irrigation activities along this river were disrupted between 2012 and 2015 due to intensive sand harvesting, which altered the riverbed and triggered conflicts among users. Since then, rehabilitation efforts – including community-led initiatives and the construction of sand dams – have helped restore water availability and enabled irrigation activities to resume (Government of Makueni County 2023; Duker *et al.* 2025).

In both countries, land and water resources are largely accessed with limited formal monitoring or coordination among users. However, important contextual differences shape irrigation decisions. In Zimbabwe, land access is mediated through traditional leadership without formal title deeds, and farmers operate in an economically volatile environment characterized by high inflation, limited access to credit, and uncertain input and output markets (Duker *et al.* 2020; Chauruka, 2025, unpublished field notes). These conditions may contribute to more cautious investment behaviour and a stronger focus on risk minimization and self-financing. In Kenya, by contrast, farmers typically have formal ownership of land along riverbanks, and relatively better access to financial services, including cooperative loans and microfinance (Government of Makueni County 2019, 2023). These conditions support greater willingness to invest and take risks. As a result, differences in land tenure issues, financial systems, and broader economic stability contribute to distinct perceptions of risk and household investment priorities across the two contexts.

2.3. Game design, development and testing

The Sand River Game was developed as a simplified representation of irrigation decision-making along sand rivers, allowing farmers to experiment with different investment strategies under changing environmental and market conditions. Detailed descriptions of the game components, rules, and scoring mechanisms are provided as supplementary material.

The game was developed through an iterative design process involving researchers and professional game designers between October 2024 and February 2025 (Table 1). The process included conceptualization, game testing and refinement stages to ensure the game realistically reflected irrigation decision-making in sand river systems. During the conceptualization stage, the game was designed with three objectives: (i) to simulate irrigation investment decision-making by smallholder farmers along sand rivers, (ii) to create a participatory platform for learning and reflection on irrigation investments and resource management, and (iii) to explore how contextual differences influence farmer decisions and outcomes. These objectives informed selection of game components, including irrigation technologies, crop production options, market conditions, and environmental risks.

The game structure was informed by the AdIP framework, which conceptualizes irrigation development as a sequence of incremental investment decisions responding to changing environmental and socio-economic conditions. Within the game, players represent farming households that make sequential decisions about irrigation

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Table 1 | Phases and stages in the design, development, testing and game play of the Sand River Game.

Phase/Stage	Objective	Main Actors	No. of Participants	Number and duration of meetings	Period
1. DESIGN AND DEVELOPMENT					
1.1 Conceptualization	Articulate the conceptual foundation of the game, including its objectives, gameplay mechanics, and its role as a tool for joint learning among farmers on irrigation investment decisions	Experts from Focus Games Limited and researchers from IHE Delft, University of Zimbabwe, and Indian Institute of Technology	7	6 × 1-hour virtual meetings	Oct 2024–Feb 2025
1.2 Practical implementation	Reflect on the game's design, playability, and its potential to support learning and dialogue on irrigation investment pathways	Same as above	7	3 × 1-hour virtual meetings	Feb 2025
2. TESTING					
2.1 Core research team	Assess consistency of modifications and alignment with intended learning and research objectives	Same as above	7	4 × 1-hour virtual meetings	Mar–Apr 2025
2.2 Farmers in south-west Zimbabwe	Evaluate farmers' understanding of the game and generate insights from their experiences to refine both the game and understanding of irrigation decision-making	Farmers from Wards 16 and 17 (Tuli River, Gwanda District)	10	1 × 3-hour in-person meeting	May 2025
2.3 A4STORE researchers	Obtain feedback on relevance and appropriateness of the game while reflecting on its potential as a learning tool across contexts	A4STORE partners (Ethiopia, Kenya, India, Mozambique, Netherlands, Niger, Zimbabwe) and Focus Games Limited	25	1 × 2-hour in-person meeting	May 2025
3. GAMEPLAY AND POST-GAME REFLECTIONS					
3.1 Zimbabwe	Simulate irrigation investment decision-making, explore trade-offs between technologies, crops, and risks, and generate farmers' reflections on irrigation investment pathways through gameplay and discussion	Farmers from Wards 16 and 17 (Tuli River, Gwanda District)	8	1 × 3-hour in-person meeting	Nov 2025
3.2 Kenya	Same as above	Farmers from Kasikeu Ward (Enguli River, Makueni County)	18	1 × 4-hour in-person meeting	Jan 2026

technologies, crop production strategies, livestock management, and complementary investments such as fencing. Detailed options for water abstraction, field irrigation, and cropping in sand river systems are provided in Appendix B.

To encourage discussion and collective reflection, the game introduces uncertainties such as climate variability, fluctuating market prices, and random events, requiring players to adapt their investment strategies across multiple simulated agricultural seasons. The game simulates irrigation investment decision-making within sand river systems by linking farmer investment choices with environmental variability, market dynamics, and shared water resources (Figure 1). Through gameplay interactions and facilitated reflection, the game creates a space for farmers to experiment with strategies and reflect on irrigation investment pathways.

The game underwent two rounds of testing prior to field implementation to assess its clarity, realism, and usability. The first test play involved members of the core research team who participated in the design process. This internal testing helped identify issues related to gameplay balance and realism, including the representation of water depletion, the influence of chance on harvest outcomes, static crop prices, and inconsistencies in asset valuation.

The second game test was conducted with irrigation farmers from Wards 16 and 17 along the Tuli River in Gwanda District, Zimbabwe. Five teams comprising of two farmers in each team participated in a three-hour

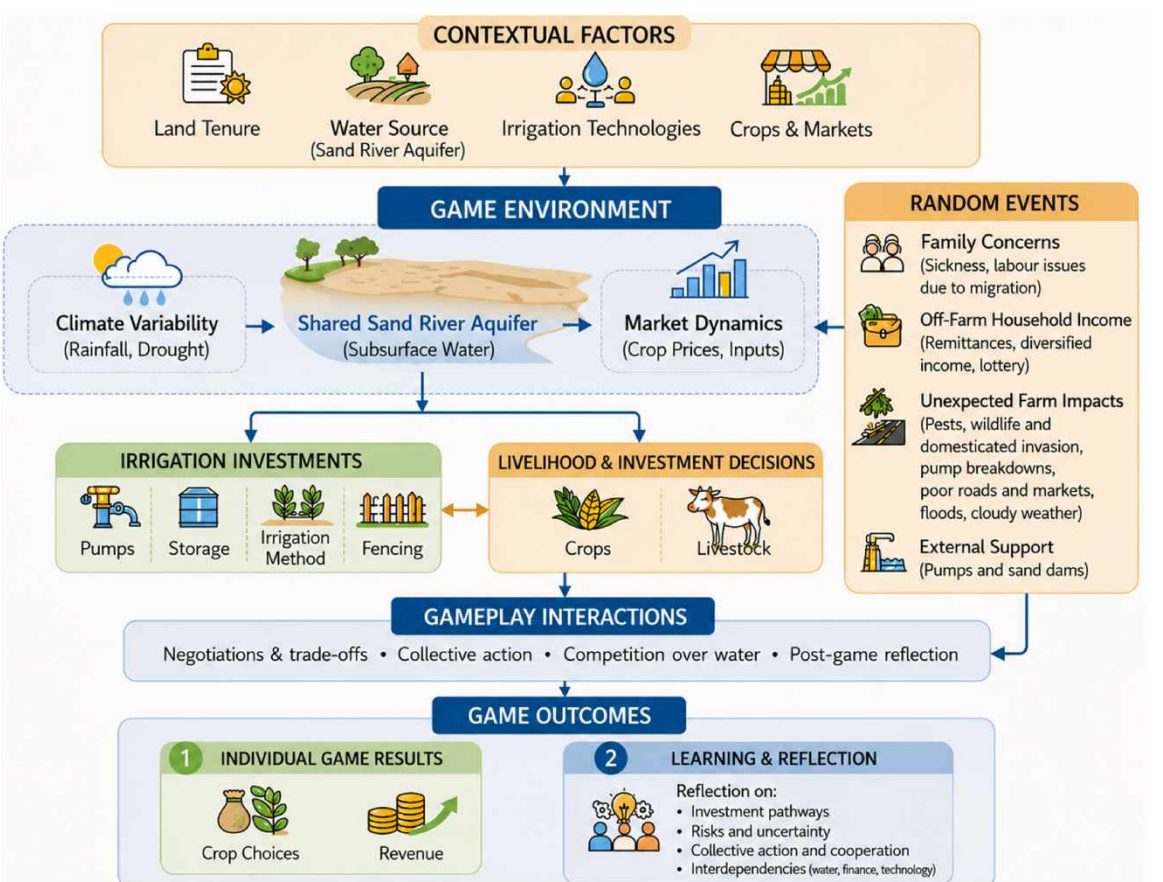


Fig. 1 | Conceptual structure of the Sand River Game.

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gameplay session including a facilitated discussion. The game tests assessed participants' understanding of the rules, perceived realism of the scenarios, and ability of the game to stimulate discussion about irrigation investment decisions. Feedback from this session informed further adjustments to event cards, gameplay flow, and facilitation procedures.

2.4. Game play sessions

In Zimbabwe, the game was played on 26 November 2025 by four teams composed of farmers from Wards 16 and 17, with each team consisting of two members. In Kenya, the game was played on 23 January 2026 by six teams composed of three farmers each from Kasikeu Ward. Across both countries, participants were purposively selected to represent diverse irrigation technologies and farming experiences and investment trajectories. Teams were used as the primary gameplay units to simulate household-level decision-making processes, reflecting how irrigation investment decisions are often negotiated within farming families rather than made by individuals in isolation. Although the number of teams differed between countries (four in Zimbabwe and six in Kenya), the study was designed as an exploratory comparative analysis rather than a statistically representative assessment. Consequently, the objective was not to compare representative farmer populations, but to examine how gameplay interactions, reflections, and irrigation investment pathways emerged under contrasting socio-environmental contexts.

The composition of the teams in both study sites is summarized in Table 2.

During gameplay, teams represented farming households managing a limited budget to make investment decisions across multiple simulated agricultural seasons. Although the game allows simulation of up to five agricultural years, gameplay sessions were limited to approximately three hours to maintain participant engagement. As a result, three simulated years were completed in Zimbabwe and two in Kenya. The financial outcomes generated should therefore be interpreted as illustrative of investment trajectories within the simulation rather than as directly comparable measures of profitability between the two case studies.

Table 2 | Composition of teams who played the game in Zimbabwe and Kenya.

Team	Gender composition	Irrigation technology and farming status in reality
Zimbabwe		
1	Two males	Diesel pump; Solar pump (adopted after pilot participation)
2	One male and one females	Hybrid (solar and petrol); Manual irrigation (buckets)
3	Two males	Petrol pump (transitioning to solar); Solar-powered system (fodder production)
4	One male and one female	Solar-powered drip system; Hybrid (solar and petrol)
Kenya		
1	Two female and one male	Manual irrigation; Petrol pump; Irrigation paused due to river channel shift
2	One male and two females	Petrol pump; Manual irrigation; Irrigation paused due to river channel shift
3	One male and two females	Solar-powered system; Petrol pump; Manual abstraction
4	One male and two females	Solar-powered system; Manual irrigation
5	One male and two females	Manual irrigation; Petrol pump with storage
6	Three females	Petrol pump with storage; Irrigation paused (health issues and river channel shift)

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2.5. Data collection and analysis

Data was collected during and immediately after gameplay sessions to capture both the decisions made by participants and the discussions and reflections generated through the game. Three main sources of data were used. First, structured game play records were used, which captured team irrigation technology investments, crop choices, financial expenditures and returns, and responses to environmental, market and/or random events. These records provided quantitative information on investment strategies and economic outcomes across teams and study sites. Second, participant observation during gameplay was used to document discussions, negotiations, and decision-making processes within teams. Observations focused particularly on how participants negotiated investment trade-offs, interpreted game events, and discussed implications for shared sand river water resources. Third, post-game reflection discussions allowed farmers to reflect on strategies used during the game, assess the realism of the scenarios, and relate gameplay experiences to real-world irrigation practices and challenges.

The collected data was analysed in relation to the study's research questions. Gameplay records were reviewed to identify patterns in investment strategies, technology adoption, crop choices, and financial outcomes across teams and study sites. Qualitative data from observations and reflection discussions were analysed using thematic analysis to identify recurring themes related to learning and experimentation during gameplay, negotiation of investment trade-offs, perceptions of shared water use, and reflections on irrigation investment pathways.

3. RESULTS

The results are organized into three main components. First, the irrigation investment pathways that emerged as teams experimented with different strategies are analysed. Second, the structure of the Sand River Game is examined to understand how it facilitated learning, dialogue, and interaction among farmers during gameplay. Finally, reflections from farmers on the usefulness of the game as a tool for exploring irrigation investment decisions are presented.

3.1. Irrigation investment pathways emerging during gameplay

3.1.1. Magnitude and timing of investments

Investment decisions during gameplay followed sequential pathways in which teams progressively invested in irrigation infrastructure, crop production, livestock, and protective infrastructure. However, the timing and magnitude of investments varied across teams and between Zimbabwe and Kenya (Figures 2 and 3).

In Zimbabwe, teams invested most heavily during the first simulated year. Annual investments ranged between approximately 49 and 112 coins in the first year. Investment levels declined in Year 2 for Teams 2, 3 and 4 while Team 1 more than doubled its investment. Investment levels stabilized at lower levels in Year 3. In Kenya, a similar pattern of front-loaded investment was observed during the two simulated years. Teams invested between 30 and 51 coins in Year 1, followed by a reduction to 10–30 coins in Year 2.

In both countries, cumulative investment pathways revealed substantial variation between teams (Figures 2 and 3). In Zimbabwe, total investments over the three simulated years ranged from 140 to 271 coins, while in Kenya cumulative investments ranged from 45 to 71 coins across the two simulated years.

3.1.2. Allocation of investments across categories

Investments during gameplay were directed towards irrigation infrastructure, crop production, and livestock, although the relative allocation of investments across these categories differed across teams and between Zimbabwe and Kenya (Figures 4 and 5).

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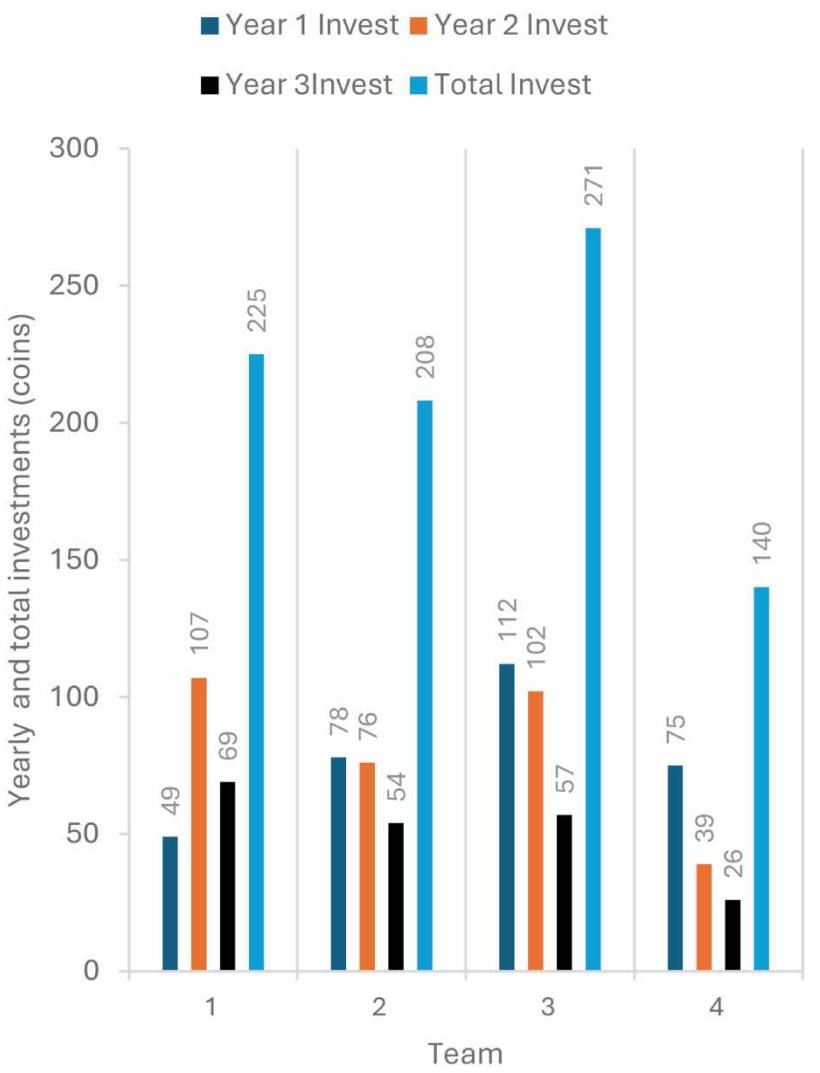


Fig. 2 | Yearly and cumulative investments by different teams in Zimbabwe over a 3-year period.

In Zimbabwe, crop production accounted for the largest share of cumulative investments across all four teams, representing between 60 and 72% of total investments. In Kenya, however, investment allocations varied more strongly across teams. While Teams 3, 4 and 5 allocated the majority of their cumulative investments to crops (60, 61 and 63% respectively), Teams 1 and 2 prioritised irrigation infrastructure with 50 and 73% respectively. Team 6 displayed the most diversified portfolio, with the majority of investments allocated to crop production, followed by irrigation infrastructure while also investing in livestock. Operational costs, including fuel for petrol pumps, are reflected within crop production investments.

In both countries, irrigation infrastructure constituted the second largest investment category. In Zimbabwe, Teams 1, 2, and 3 allocated approximately a quarter of their cumulative investments to irrigation infrastructure,

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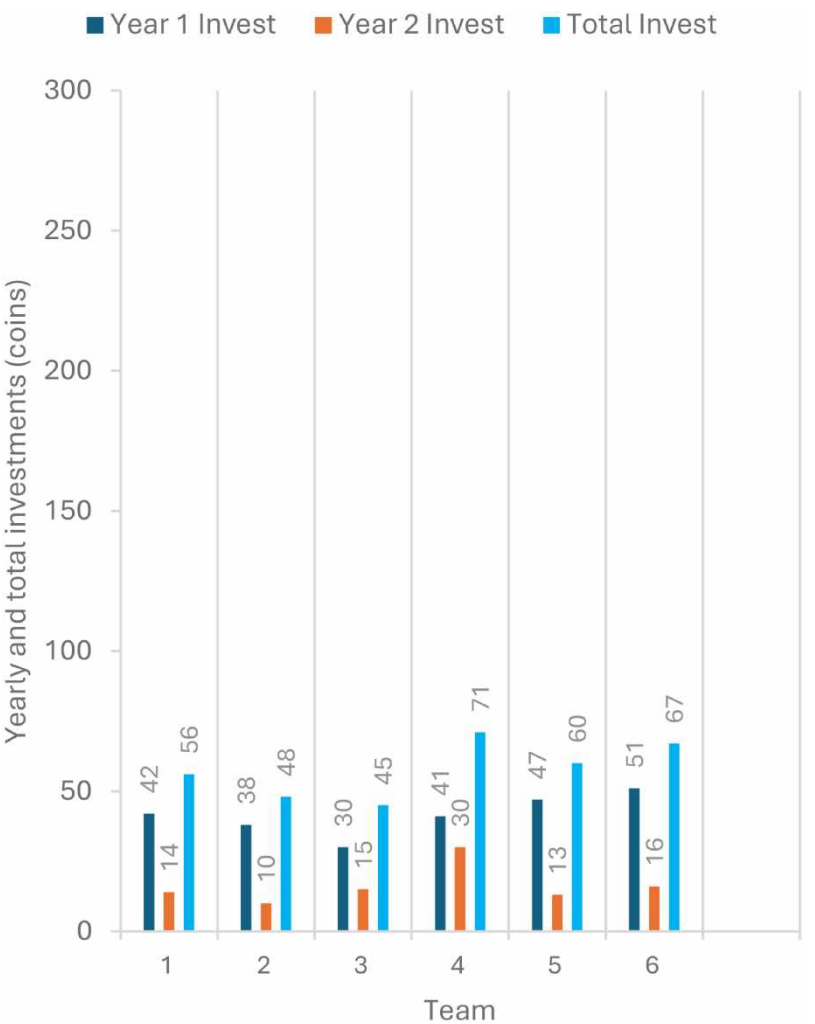


Fig. 3 | Yearly and total investment by different teams in Kenya over a 2-year period.

while Team 4 allocated a considerably smaller share. In Kenya, infrastructure investments were above 50% for half of the teams while for the other half, they were below 50%.

Investments in livestock represented the smallest share of cumulative investment portfolios in both countries, although their role differed between contexts. In Zimbabwe, livestock accounted for between 7 and 17% of cumulative investments, with all teams investing in livestock, while in Kenya, only Team 6 invested a small share towards livestock as part of a diversified investment strategy alongside irrigation and crop production.

The timing of investments across categories also reflected the sequential nature of irrigation development. In both countries, irrigation infrastructure dominated annual investments in Year 1 of gameplay. In Zimbabwe, Teams 1 and 2 allocated 64% of their annual investments to irrigation infrastructure across the first two years, with Team 2 concentrating these investments primarily in Year 1. Investment levels towards irrigation infrastructure declined by Year 3, with only Team 1 investing 5%. Similarly, in Kenya irrigation infrastructure

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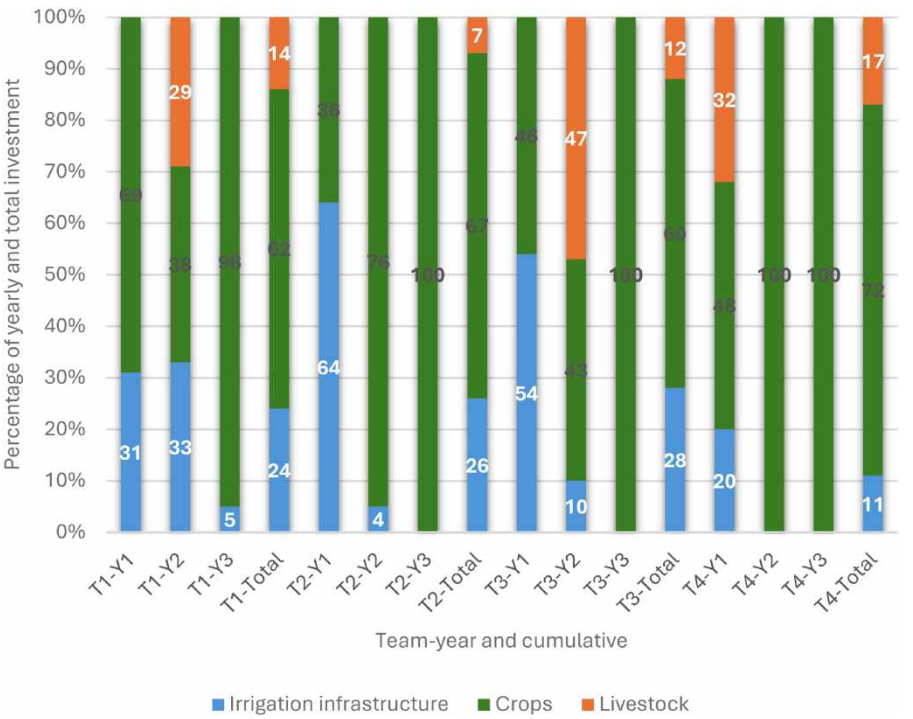


Fig. 4 | Proportional allocation of annual and total investments by different teams in Zimbabwe over a 3-year period.

accounted for above 50% of investments in Year 1 for two thirds of the teams, including 92% of annual investments for Team 2, 81% for Team 5, 67% for Team 1, and 60% for Team 3.

In subsequent years, investment patterns shifted more towards crop production for both country cases, with all teams in Zimbabwe investing almost 100% towards crop production in the third year of gameplay. In Kenya, a similar shift occurred in Year 2, when all teams except Team 3 allocated all their annual investments towards crop production.

The concentration of infrastructure investment in the early stages of gameplay is further reflected in the timing of specific infrastructure acquisitions. However, this early clustering of investments should be interpreted with caution, as it may partly reflect the structure of the game, particularly the early availability and framing of irrigation technologies, rather than fully representing real-world decision-making dynamics. In both countries, all teams acquired irrigation pumps during the first year of gameplay. In Zimbabwe, Teams 1 and 4 opted for petrol-powered pumps, while Teams 2 and 3 adopted hybrid strategies combining petrol and solar pumps after selling livestock to finance these investments. Similarly, in Kenya, Teams 1, 3, 4, and 6 invested in petrol-powered pumps, while Teams 2 and 5 adopted solar-powered systems, illustrating two distinct technological pathways: fuel-based pumping systems with lower initial investment but higher operating costs, and solar-powered systems with higher upfront costs but lower operating expenses.

Complementary infrastructure investments emerged gradually and often in response to risks encountered during gameplay. In Zimbabwe, Team 3 invested in a drip irrigation system during Year 1 to improve water-use efficiency, while investments in fencing and storage infrastructure were generally delayed until Year 2 after teams experienced crop losses due to events such as livestock intrusion, wildlife damage and loss of crops for

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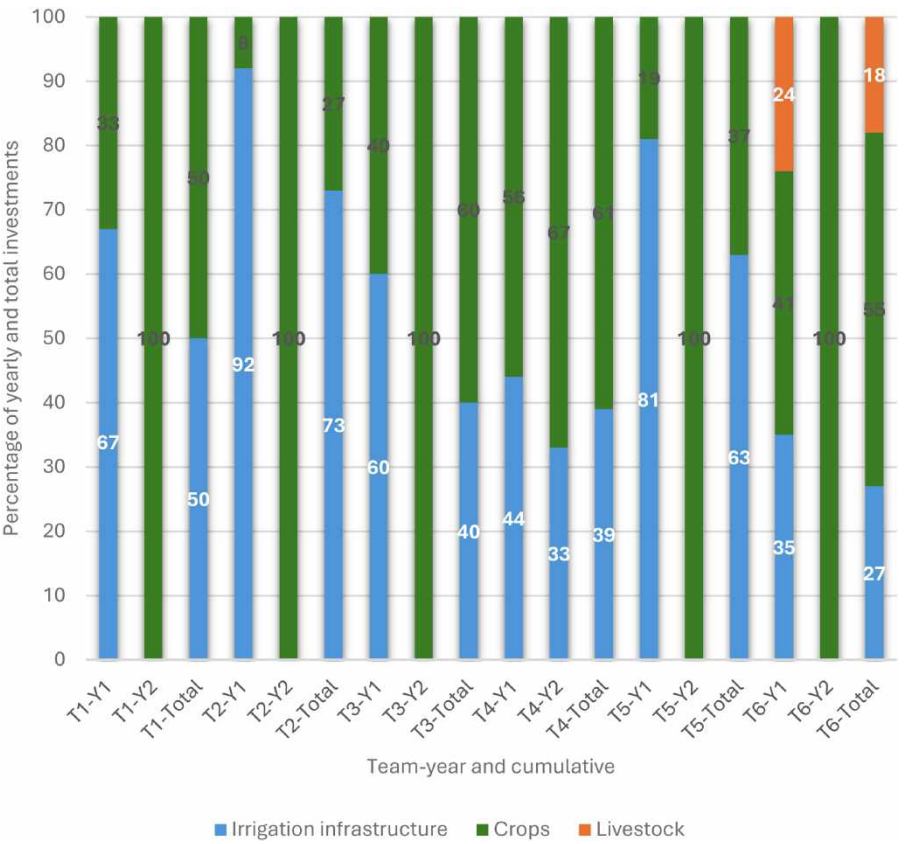


Fig. 5 | Proportional allocation of annual and total investments by different teams in Kenya over a 2-year period.

teams with no storage. Similar adaptive responses were observed in Kenya, where Team 1, 5 and 6 invested in fencing during Year 1, while Team 2, 3, and 4 introduced fencing later after recognizing exposure to grazing risks.

Event cards further influenced investment capacity and decisions in both countries. For example, in Zimbabwe Team 1 received a solar pump through a Non-Governmental Organization (NGO) supported event card, strengthening its irrigation capacity, while in Kenya a similar event allowed Team 1 to acquire an additional solar pump. Event cards also introduced operational costs and risks associated with irrigation technologies. In Zimbabwe, pump maintenance costs were incurred in Year 3, while in Kenya external shocks such as weather variability influenced the performance of irrigation technologies. Teams that invested in diversified infrastructure systems were better able to maintain productivity despite shocks introduced through event cards. For example, Team 3 in Zimbabwe, which invested in a hybrid pumping system combining petrol and solar pumps together with drip irrigation and later a storage tank, was able to maintain crop production despite adverse events such as bad weather. In Kenya, Team 1 strengthened its irrigation capacity by combining petrol pumping with storage infrastructure and an additional solar pump obtained through an NGO-supported event card. Team 4 supplemented its irrigation system with a storage tank in the second year. These combinations of infrastructure investments helped buffer the effects of environmental and operational shocks.

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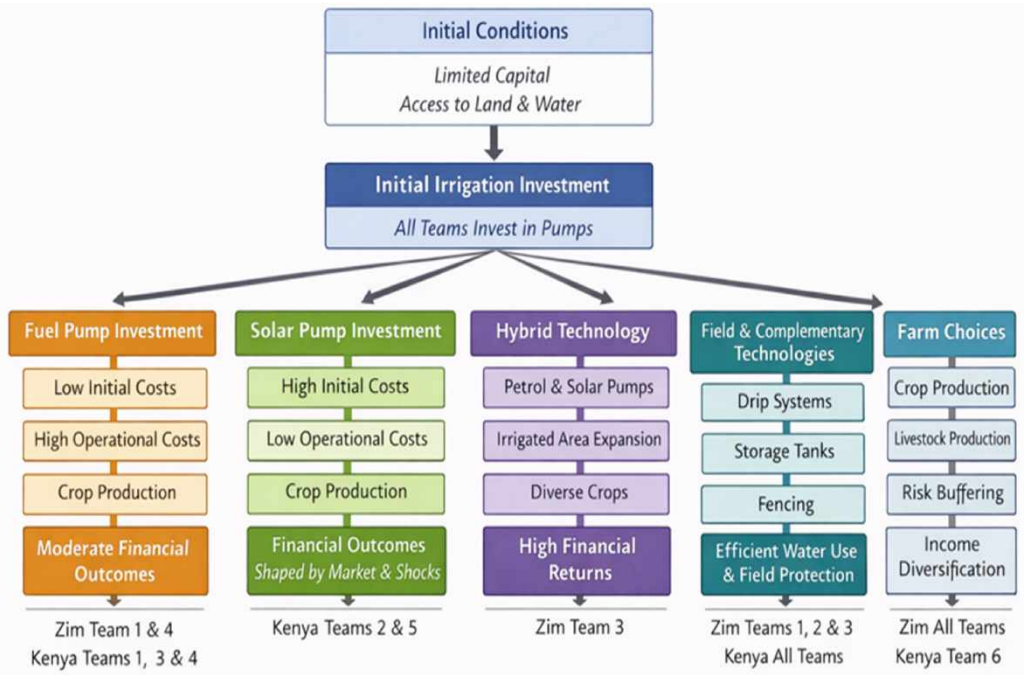


Fig. 6 | Emergingirrigation investment pathways in Kenya and Zimbabwe during Sand River Game simulations.

3.1.3. Drivers and adaptive responses shaping irrigation investment pathways

Analysis of gameplay outcomes revealed the key factors that shaped investment pathways across teams and countries. These factors included environmental variability, market conditions, technology choices and performance, and external support through events (Figure 6).

Environmental variability affected crop production in both countries. In Zimbabwe, a below-average rainy season event card reduced sand river recharge and resulted in water shortages affecting irrigated plots. Teams 1 and 2 each lost two irrigated fields, while Teams 3 and 4 lost one field each, reducing crop production during that season. In Kenya, irrigation areas remained relatively small and a sand dam event card increased water availability, preventing water-related crop losses. However, grazing livestock damaged crops on unfenced plots belonging to Teams 2, 3, and 4, which prompted these teams to invest in fencing later in the gameplay to protect their fields from further losses.

Market conditions and crop profitability significantly influenced decisions. In Zimbabwe, Teams 1 and 2 invested in butternut production in Year 1, which resulted in financial losses due to lack of local market and collective marketing arrangement. Following these outcomes, the teams shifted subsequent investments towards crops with more reliable local markets such as tomatoes, maize, and leafy vegetables. Similar experiences were observed in Kenya where Teams 3 and 4 also encountered financial losses after investing in butternut. In addition, Team 5 invested in fruit trees, anticipating higher long-term returns. However, the high establishment and maintenance costs of the fruit trees exceeded the market prices, resulting in lower profitability than expected. These experiences influenced subsequent investment decisions, with teams prioritizing crops such as maize, leafy vegetables and tomatoes. Collective marketing opportunities also influenced crop choices and investment strategies, particularly in Zimbabwe. At the start of Year 3, Teams 2 and 3 joined the collective

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market, which offered higher prices compared to local markets. Anticipating these higher returns, both teams expanded production of butternuts, a crop that previously had no local market in earlier rounds of the game. Team 3 planted three fields of butternuts, while Team 2 planted two fields, strategically targeting the collective market where butternuts could only be sold, and at higher prices. In contrast, no team in Kenya game joined a collective market, and farmers relied solely on local market opportunities.

Irrigation technology performance and operational costs further guided investment decisions. Teams using petrol pumps faced recurring fuel costs, while solar-powered systems were sensitive to cloudy weather conditions. Hybrid strategies combining petrol and solar pumps allowed teams in both countries to balance reliability and operating costs. For example, in Zimbabwe Team 3, which adopted a hybrid system and added a storage tank, maintained productivity despite cloudy weather events, whereas teams with single-mode technologies had less flexibility. Similarly, in Kenya, Team 1, which invested in a petrol pump, storage tank, and fencing and later received a solar pump via an NGO event card, was able to sustain irrigation operations during adverse conditions.

External support introduced through event cards also influenced investment pathways. In Zimbabwe, Team 1 received a solar pump through an NGO-supported event card in Year 2, increasing their irrigation capacity by enabling the team to combine solar and petrol pumping. Zimbabwe also experienced infrastructure-based external support when Team 4 drew an event card indicating that an NGO proposed the construction of a sand dam, which added three water cards back into the river system and increased the pool of available water for irrigation. In Kenya, external support similarly appeared through water infrastructure when Team 6 drew an event card indicating that an NGO built a sand dam, adding three water cards back into the river system. Thus, rather than directly supporting a specific team, this intervention increased water availability for all players. These events illustrate how external interventions in the game influenced irrigation outcomes either by directly increasing farmers' irrigation assets, as in the case of the solar pump in Zimbabwe, or by enhancing shared water resources, as observed through sand dam construction in both Zimbabwe and Kenya. These patterns were consistent with participants' real-world irrigation practices and farming situations (Table 2). However, some random events, particularly those involving windfall gains such as lotteries or externally provided assets, were perceived by participants as less realistic. They occasionally disrupted strategic planning by introducing outcomes that appeared overly dependent on chance. While these events were intended to simulate uncertainty and external interventions commonly encountered in development contexts, future versions of the game could calibrate event probabilities more closely to observed real-world frequencies. The events were, however, important as they stimulated discussion on uncertainty, unequal access to external support, and how sudden opportunities or shocks can alter irrigation investment trajectories.

Figure 6 summarizes these dynamics: the first three columns depict investments in water abstraction technologies, the fourth column shows field and complementary technologies, and the fifth column represents farm-level choices including crop and livestock production, income diversification, and risk management strategies.

3.1.4. Financial outcomes of investment pathways

Differences in investment pathways resulted in varying financial outcomes across teams and across the case studies.

By the end of Year 3 in Zimbabwe, Team 3 accumulated the most revenue (137 coins), followed by Team 2 (91 coins), Team 4 (49 coins), and Team 1 (43 coins). Team 3's success was due to an early expansion of irrigated area, rapid transition to solar-powered drip irrigation, strategic use of collective marketing for high-value crops, and investment in livestock. Team 2 performed moderately due to investments in fencing and drip irrigation adoption. It was, however, its perfomance was constrained by high petrol pump costs and poor markets.

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By the end of Year 2 in Kenya, Team 6 recorded the highest total revenue (61 coins), followed by Team 1 (55 coins), Team 5 (52 coins), Team 4 (41 coins), Team 3 (37 coins), and Team 2 (27 coins). Team 6's strong performance was driven by diversification into high-value crops (leafy vegetables and tomatoes), investment in livestock, protective fencing, and loan utilisation. A 20-coin lottery event in Year 1 also boosted its total revenue. Team 1 accumulated 55 coins through maize, tomatoes, and leafy vegetables, supported by fencing installed in Year 1 and a free planting event. Team 5 generated revenue through diversified cropping and loan use. However, market limitations on butternut sales and negative event cards reduced income in certain seasons.

Lower-performing teams were also observed across both case studies. In Zimbabwe, Team 1 (43 coins) and Team 4 (49 coins) recorded the lowest revenues. These teams experienced crop losses associated with event cards, including pest outbreaks, grazing events, and limited market access for certain crops such as butternut. In Kenya, Team 2 (27 coins) and Team 3 (37 coins) recorded the lowest revenues. Team 2 faced early financial constraints, including partial payment of pump maintenance costs and reliance on loans, as well as crop losses linked to event cards such as floods and medical expenses. Team 3 experienced repeated crop losses associated with events such as wildlife damage and floods and had several seasons with no harvest.

The financial outcomes are based on a simulation period of two to three years. Longer-term dynamics such as infrastructure replacement, resource depletion, and cumulative risk exposure were not represented within the gameplay. Because gameplay durations differed between the Zimbabwe and Kenya sessions, the financial outcomes from the gameplay should not be interpreted as directly comparable indicators of investment performance across countries.

3.2. How the game facilitated learning and reflection

The Sand River Game created a structured environment in which farmers experimented with irrigation investment strategies, responded to environmental and market shocks, and reflected collectively on the consequences of their decisions. Learning occurred through iterative gameplay, team discussions, and facilitated reflection sessions after the simulation.

3.2.1. Experiential learning

Experiential learning during gameplay was reflected in changes in irrigation investments, crop choices, and farm management decisions across successive rounds following environmental and random events, market outcomes, and operational constraints.

Crop losses caused by wildlife incursions and livestock entering unfenced fields influenced investments in protective infrastructure. For example, in Kenya, Teams 3 and 4 lost their irrigated fields during the first dry season due to an elephant invasion event and grazing cattle event respectively, prompting investment in farm fencing the following season.

Market outcomes also shaped crop decisions. In Zimbabwe, Team 2 planted butternut in Year 1 when the crop had no local market, resulting in no income. When a collective marketing opportunity was introduced later in the game, Teams 2 and 3 joined and expanded butternut production to gain from the higher collective market prices.

Changes in irrigation technology occurred during gameplay. In Zimbabwe, Team 2 initially invested in a petrol pump to enable irrigation during early dry seasons. After experiencing high fuel costs during successive planting seasons, the team later invested in a solar pump. It retained the petrol pump as backup, resulting in a hybrid pumping arrangement combining solar and petrol pump.

Solar irrigation investments also led to complementary infrastructure investments. For example, in Kenya, Team 2 invested in solar system with no storage. The team lost an irrigation field due to a 'cloudy weather'

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event which reduced solar energy and limited pumping capacity. This prompted an investment in a water storage tank to store pumped water and stabilize irrigation water supply.

Across successive gameplay rounds, teams modified irrigation technologies, infrastructure investments, and crop production strategies in later seasons after experiencing crop losses, market constraints, and operational costs in earlier rounds.

3.2.2. Dialogue and negotiation

The game required teams to make collective decisions regarding irrigation investments, crop production, and financial management. These decisions were negotiated within teams, creating opportunities for dialogue about farming strategies and risk management.

In Kenya, discussions frequently emerged around whether to prioritize irrigation infrastructure, or crop production. For instance, when Team 2 faced a financial shock after being required to pay petrol pump maintenance costs in Year 1, team members debated whether to invest further in crop production or focus on income-generating activities such as livestock. Similarly, Team 4 discussed whether to expand irrigation investments through borrowing, where in Year 2, the team took a revolving fund loan to expand production, investing in a storage tank and crop production. Dialogue also emerged around the role of markets. When Team 6 planted butternut, the absence of a local market resulted in no income from the crop, prompting discussion among participants about market access and crop selection. These team-level negotiations encouraged participants to share experiences and debate different approaches to irrigation development.

In Zimbabwe, team discussions similarly focused on balancing investments in irrigation technology, crops, and livestock. In Year 1, Teams 2 and 3 debated whether to purchase a petrol pump, or a solar pump. They eventually sold their initial livestock to buy both pumps. Discussions around risk management were prominent, particularly when teams faced adverse events such as 'grazing cattle' or floods. Market considerations were central to team negotiations, with decisions about which crops to grow influenced by potential local or collective market opportunities. Teams reflected on previous market outcomes to guide planting decisions. Team 2 and 3, joined collective marketing to secure higher prices for butternuts.

Across both countries, these dialogues reflected trade-offs between irrigation technology costs, operational efficiency, crop selection, and market access under environmental and financial shocks. The structured negotiation enabled teams to experiment with different irrigation pathways, respond to events, and make decisions that directly influenced financial outcomes. In both countries, teams that engaged in collective marketing and diversified their investments were able to achieve higher returns, while teams focusing narrowly on single crops or avoiding loans faced lower or no harvest income. These differences in decision-making were consistent with participants' real-world irrigation practices and resource contexts (Table 2). However, participation within teams was not always evenly distributed. In some cases, more experienced or outspoken participants dominated decision-making, potentially limiting the contributions of less experienced farmers. These intra-team dynamics may have influenced the investment strategies adopted and the nature of the discussions observed.

3.2.3. Reflection on risks and trade-offs

The game exposed participants to multiple sources of risk, including environmental shocks, financial constraints, and market fluctuations. These required teams to evaluate trade-offs between different investment options.

Environmental risks played a particularly important role. For example, in Kenya, in Year 2, all teams lost crop fields due to a flood event, while Team 3 experienced reduces crop prices in Year 2 due to a flooded market event. Similarly, in Zimbabwe, all teams lost crop fields in Year 3 due to a flood event.

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Financial shocks also shaped decisions. In Kenya, Team 5 faced a hospital bill, prompting the team to access a revolving fund loan to meet the expense. While the loan enabled continued production, it also required subsequent repayments, affecting later investment decisions. Such events prompted participants to reflect on the risks associated with irrigation investments and the importance of diversification and financial planning.

While the game stimulated rich discussion on irrigation investment decisions, these interactions were limited to irrigating farmers. As a result, perspectives from other stakeholders relevant to sand river systems, such as domestic water users, livestock owners, and water authorities, were not represented. This may have contributed to the limited attention to broader water governance and allocation issues during gameplay, which only emerged more strongly during post-game reflections.

3.3. Farmer reflections on the game

3.3.1. Zimbabwe

After the game, farmers reflected on their experiences and highlighted several key insights. Participants noted that the game closely reflected real-world opportunities and challenges facing farmers. There was a strong recognition of the importance of protective investments, particularly fencing, to reduce crop losses from domestic animals and wildlife. Farmers also emphasized the importance of appropriate crop selection in relation to market availability, noting that collective marketing was critical for certain crops.

Livestock was identified as an important asset for enhancing resilience, contributing to liquidity and enabling asset accumulation through sales. Farmers further highlighted the importance of knowing when to destock, as livestock market prices varied across seasons. Petrol pumps were widely acknowledged as relatively affordable to acquire but costly to operate, leading to discussions on the advantages of transitioning to solar-powered systems.

Participants also noted the risks associated with cultivating large areas during periods of water scarcity without reliable information on river conditions. In this context, farmers emphasized the value of shared water monitoring systems and expressed enthusiasm for the planned installation of piezometers as part of the *Smallholder farming families:* **A**dapt **A**frican **A**lluvial **A**quifers to **S**trengthen **T**heir **O**wn **R**esilience (A4STORE project). Additionally, one farmer reported that participating in game testing motivated him to sell two cows and invest in a solar-powered irrigation system.

3.3.2. Kenya

Farmers in Kenya similarly described the game as a realistic representation of the challenges encountered in practice. They emphasized the importance of prioritizing crops with reliable local and external markets. They noted that planting multiple crops helped to spread risk and improve income stability. Careful planning was highlighted as critical, including decisions related to timing, crop selection, irrigation infrastructure, and protective measures such as fencing to reduce losses from floods, pests, and wildlife.

Participants recognized the role of calculated risk-taking in agricultural investment, including the use of loans to expand production or invest in infrastructure. At the same time, some noted that avoiding loans limited opportunities for growth. Farmers emphasized the importance of maintaining sufficient capital and savings, developing contingency plans to cope with unexpected shocks, and diversifying livelihoods, particularly through livestock, to strengthen income streams.

Learning from experience was identified as an important aspect of decision-making. Participants highlighted the need to reflect on past actions, adapt to challenges, and adjust investment strategies over time. Collective marketing was also seen as important for accessing better prices for high-value crops. Some participants

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expressed intentions to take loans from the Kasikeu Service and Financial Association (KSFA) to invest in irrigation equipment, reflecting increased willingness to engage with financial services.

Finally, a number of participants noted that the use of coins as a currency was somewhat abstract and not always intuitive when relating costs and returns to real-world values. Attempts were made to equate coins to local currency, suggesting that closer alignment with familiar monetary values could enhance the game′realism and participant engagement.

4. DISCUSSION

This study examined how serious gaming can support learning and reflection among farmers regarding irrigation investment decisions in sand river systems and how gameplay can reveal potential irrigation investment pathways. The discussion situates these findings within broader debates on decentralised irrigation development and the role of participatory tools in supporting irrigation planning and water governance.

4.1. Irrigation investment pathways in FLID

The findings provide insights into how farmers explore and adjust irrigation investments under conditions of uncertainty. Across gameplay rounds, teams modified irrigation technologies and crop choices in response to operational costs, market constraints, and random events experienced in earlier rounds. These sequential adjustments reflect the incremental and adaptive nature of investment decisions commonly observed in FLID.

FLID research highlights that irrigation development among smallholder farmers rarely follows a predetermined or linear plan but instead emerges through gradual investments and experimentation as farmers respond to opportunities and constraints (Woodhouse *et al*. 2017; de Bont *et al*. 2019). The investment patterns observed during gameplay reflect similar dynamics.

Technology choices also evolved as teams encountered operational constraints. Teams that initially relied on petrol pumps adjusted their strategies after high fuel expenditures by introducing solar-powered systems or hybrid pumping arrangements. These transitions highlight how operational costs can shape technology adoption trajectories. Rather than abandoning irrigation investments altogether, teams reconfigured their technological choices in order to maintain production while reducing recurrent costs.

The findings further suggest that irrigation investment pathways are strongly path-dependent, with early investment decisions shaping the range of future opportunities available to farmers. Initial choices regarding irrigation technologies, access to finance, and crop selection influenced subsequent investment flexibility, responses to operational costs, and the capacity to respond to shocks. These findings have important implications for irrigation policy in SSA, where irrigation development interventions often assume linear planning trajectories and standardized infrastructure models (McCarthy 2022). The findings suggest that smallholder irrigation development may be better supported through flexible financing mechanisms, adaptive support systems, and investment approaches that recognize incremental and evolving farmer strategies.

Investment behaviour also reflected broader livelihood strategies. In Zimbabwe, livestock functioned not only as a productive asset but also as a financial buffer that could be mobilized to finance irrigation-related investments such as fencing or irrigation equipment. This mirrors farming practices in Zimbabwe (Scoones 1992; Maravanyika & Chimwai 2025). It was clear that irrigation investments were embedded within wider livelihood strategies and cannot be understood solely as technical decisions.

The findings show that no single irrigation development pathway emerged during the game sessions. Teams achieved positive outcomes through different combinations of investments in irrigation technologies, crops, livestock, and market strategies. This diversity of strategies reflects the heterogeneity of farmer-led irrigation systems.

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Farmers pursue multiple and evolving pathways depending on their resource endowments, risk preferences, and opportunities and constraints they encounter (Duker *et al.* 2023; Liebrand *et al.* 2023; Tsiko *et al.* 2024).

Analysis of gameplay interactions and participant discussions indicated that structural and environmental factors, particularly access to finance, land tenure, and water availability, shaped how farmers engaged with investment decisions and the investment pathways they explored. In Zimbabwe, participants generally cultivated multiple fields during the simulation, reflecting both the historical reliability of the Tuli River and the absence of competition for water, as significant episodes of river drying occurred decades ago, particularly in the 1970s, allowing farmers to perceive a lower risk of water shortages. Their notable reluctance to take loans – shaped by experiences of Zimbabwe's decades of macroeconomic instability, high inflation, and high interest rates – reinforced cautious, incremental investments using internally generated resources, consistent with FLID principles (Duker *et al.* 2023). In contrast, Kenyan participants planted fewer fields, reflecting concerns about limited water availability along the Enguli River, where sand depths have not fully recovered from intensive sand harvesting and dry-season shortages can further constrain supply. Lived experiences of relatively accessible and stable rural finance systems, including cooperative loans and microfinance schemes, increased willingness to take risks through credit. Farmers also benefit from formalized land tenure systems where they own the land along the river, providing an incentive to invest more aggressively in irrigation. These economic and institutional conditions facilitated reliance on start-up capital and loans to finance irrigation investments.

In both Zimbabwe and Kenya, irrigation investment pathways are not shaped solely by hydrological opportunities but are strongly mediated by broader macroeconomic and institutional conditions. In Zimbabwe, macroeconomic instability, inflation, and insecure land tenure conditions reinforced cautious and incremental investment behaviour centered on self-financing and livestock accumulation. In contrast, Kenyan farmers operated within relatively more stable rural finance systems and formalized land tenure arrangements, conditions that encouraged greater willingness to use loans and pursue more aggressive irrigation expansion. These findings reinforce arguments within FLID literature that irrigation investment decisions are embedded within broader political-economic systems rather than being purely technical responses to water availability (Woodhouse *et al.* 2017; de Bont *et al.* 2019; Duker *et al.* 2023).

Together, these patterns illustrate a key principle of the AdIP framework: early decisions, contextual conditions, and resource availability shape both immediate investment choices and the future pathways that remain open or closed to farmers (Prasad *et al.* 2023, 2024).

While the observed investment pathways suggest incremental and adaptive decision-making, it is important to acknowledge that some of these dynamics are shaped by the structure and rules of the game itself. Certain investment options and risks were embedded in the game design, which may have influenced the sequence and timing of decisions. For instance, while investments such as fencing were introduced in response to game events, farmers in reality may anticipate such risks and prioritize these investments earlier. As such, the findings should not be interpreted as direct representations of actual investment sequences, but rather as insights into how farmers reason about investment choices, evaluate trade-offs, and weigh risks and benefits within a structured decision-making environment. In addition, the sequencing of investments may reflect how risks and options were presented within the game rather than fully representing real-world anticipatory behaviour.

4.2. Exploring AdIP through serious gaming

The results also contribute to discussions on AdIP, which emphasize the need for flexible and context-specific approaches to irrigation development (Prasad *et al.* 2023). While AdIP has so far been presented as a conceptual framework, the Sand River Game provided a practical setting in which farmers could experiment with different irrigation development trajectories and observe their consequences over time.

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Through the simulation of multiple farming seasons, the game enabled participants to explore how early investments influenced later opportunities and constraints. For example, decisions to invest in irrigation infrastructure, fencing, or particular crop types affected teams' financial resources and production possibilities in subsequent rounds. Such sequencing of decisions mirrors the logic of adaptive pathways, where early choices shape the range of future options available to farmers. The game therefore operationalised the AdIP framework within a participatory setting by enabling farmers to visualise how sequential investment decisions create branching development trajectories over time. Rather than treating irrigation development as a fixed infrastructure pathway, the gameplay demonstrated how farmers continuously evaluate flexibility, risks, and future options under uncertain conditions.

The simulation enabled participants to explore trade-offs between alternative irrigation investment strategies and observe how different choices influenced financial outcomes, resource use, and resilience to environmental and market shocks. Teams faced decisions between allocating resources to irrigation infrastructure, expanding cropped area, investing in livestock, or managing risks through protective measures such as fencing. Because the game incorporated uncertain events, participants were required to reconsider strategies and adjust their investments accordingly.

The game therefore functioned as a tool for exploring potential irrigation development trajectories rather than prescribing a single optimal strategy. By enabling farmers to test alternative strategies in a low-risk environment, the simulation provided insights into how AdIP may unfold.

4.3. Serious gaming as a tool for dialogue and reflection

Beyond individual decision-making, the Sand River Game created a structured platform for dialogue among participants about irrigation development and resource management. This is consistent with the use of serious gaming as a participatory tool for exploring complex socio-ecological systems because it enables stakeholders to interact with simplified representations of real-world dynamics while reflecting on the consequences of their decisions (Speelman *et al.* 2014; Barreteau *et al.* 2021).

In this study, the board-based simulation brought together multiple dimensions influencing irrigation development. By situating teams within comparable biophysical and economic conditions, the game provided a shared reference framework through which participants could compare strategies and outcomes – an important feature for collective learning in FLID, where decisions are typically decentralised and experiential (Veldwisch *et al.* 2019).

Facilitation played a critical role in this process. Unlike digital simulations where interactions are embedded within software, board-based games depend heavily on guided facilitation to maintain coherence and stimulate reflection. The facilitator introduced events, mediated loan mechanisms, and ensured that participants understood the consequences of their decisions. This aligns with literature emphasizing that the effectiveness of non-digital serious games depends not only on game mechanics but also on structured facilitation processes that guide interpretation and learning (Rodela & Speelman 2023).

The implementation of the game in Zimbabwe and Kenya further demonstrated how local contexts shape irrigation investment strategies and the range of pathways considered by farmers. In Zimbabwe, teams more frequently incorporated livestock into their investment strategies and relied heavily on crop revenues to finance irrigation-related expenditures. In Kenya, teams relied more on start-up capital and loans to finance irrigation investments. These differences reflect broader economic and institutional contexts, reinforcing the argument that FLID is embedded within political and economic structures rather than being purely a technical process. At the same time, choices during gameplay were often dominated by short-term profitability considerations. Discussions mainly focused on maximising financial returns within the limited timeframe of the game, while longer-

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term sustainability concerns such as groundwater depletion or collective water governance received less attention. Although post-game reflections created opportunities to address these broader issues, this finding suggests that integrating structured reflection pauses during gameplay may strengthen the capacity of serious games to stimulate deeper discussions about sustainability and resource governance.

The findings also suggest that serious games can function as anticipatory governance tools by creating opportunities for stakeholders to discuss potential resource conflicts, investment trade-offs, and sustainability concerns before these materialise within real irrigation systems. In this way, serious gaming may support more reflective and participatory approaches to irrigation planning and resource management within complex socio-ecological systems (Speelman *et al.* 2014; Barreteau *et al*. 2021).

These findings also highlight the potential of serious games to complement other research approaches. While not a substitute for field-based observation or longitudinal studies, serious games provide a structured yet flexible platform to explore decision-making processes, test responses to hypothetical scenarios, and facilitate reflection among participants. When combined with empirical field data, they can enhance understanding of how farmers navigate complex and uncertain irrigation investment environments. By simulating interactions between irrigation technologies, environmental risks, and market conditions, the Sand River Game provided a simplified but interactive representation of the complex dynamics shaping irrigation development in sand river systems. The simulation created opportunities for farmers to experiment with alternative investment strategies and to observe how these strategies influenced outcomes over multiple seasons – an opportunity that is often limited in real farming contexts where financial risks constrain trial and error.

5. CONCLUSIONS

This study examined how a simulation-based serious game can support learning and reflection among farmers regarding irrigation investment decisions in sand river systems and how gameplay can reveal potential irrigation investment pathways. The design and iterative development of the Sand River Game created a simplified but realistic representation of the technological, environmental, and financial decisions that farmers face when investing in irrigation. Participants perceived the game as reflecting real farming situations, which facilitated engagement and discussion during gameplay and post-game reflection sessions. Across teams in both Zimbabwe and Kenya, farmers tended to make incremental investments while carefully weighing risks related to irrigation technologies, operational costs, market access, and environmental disturbances. These patterns closely resemble the adaptive and stepwise investment behaviour described in the FLID literature and reflect the core principles of AdIP, where irrigation development unfolds through sequential decisions made under conditions of uncertainty. The findings contribute to the literature on FLID and AdIP by showing how farmers explore investment strategies, respond to risks, and adjust decisions over time within a simulated environment.

The game also provided a semi-structured tool for dialogue about irrigation investments and their consequences. Discussions during gameplay primarily focused on production decisions, profitability, and strategies for managing operational risks such as fuel costs, crop losses, and market fluctuations. These discussions helped reveal how farmers evaluate trade-offs between different investment options and how they adjust strategies following losses or unexpected events. Other issues that are central to FLID and AdIP, such as long-term resource sustainability, collective water governance, and potential conflicts over shared resources, emerged during post-game reflection sessions.

Broadening stakeholder participation and incorporating structured reflection points during gameplay could strengthen the potential of serious games to support dialogue on governance and collective management challenges associated with sand river irrigation systems.

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The findings demonstrate that serious gaming can provide a valuable participatory tool for exploring irrigation investment decisions and stimulating dialogue among farmers about the opportunities and risks associated with irrigation development. By enabling farmers to experiment with investment strategies and observe their consequences in a low-risk environment, the Sand River Game offers a practical approach for grounding discussions of FLID and AdIP. When combined with longer-term field research and participatory engagement processes, such tools can support more informed decision-making and contribute to the development of irrigation pathways that are both productive and resilient in sand river systems (Brauner, *et al.* 2025).

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ETHICS STATEMENT

Free and informed consent of all participants was obtained prior to their participation in the study. The study involved irrigating farmers participating in gameplay sessions, observations, and post-game reflection discussions in Zimbabwe and Kenya.

DATA AVAILABILITY STATEMENT

Data cannot be made publicly available; readers should contact the corresponding author for details.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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